

Attrition Case Study

Executive Summary

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In this case study, the attrition dataset was analyzed. The case study task was to identify the top 3 factors affecting attrition. Additionally, the case study task was to build a model to predict attrition and be able to measure the cost or savings impact of the model. There were no null values for the attrition dataset. The only data wrangling required for this dataset was changing the categorical variables into factors. The main variables used in the attrition case study analysis was: job involvement, attrition, monthly income, and overtime. Job involvement was a categorical variable with a score of 1 being low to 4 being very high. The categorical variable attrition indicated whether the employee left the company (Yes) or not (No). The numerical variable monthly income was the employee's monthly salary. The categorical variable overtime, indicated whether the employee worked overtime (Yes) or not (No).

T-tests, KNN models, and Naive Bayes models were used to determine the top 3 factors of attrition. After analysis, the top 3 factors affecting attrition were monthly income, job involvement, and working overtime. Age was another important factor for attrition. The Naive Bayes Theorem using a threshold of 0.2 and incorporating the variables: MonthlyIncome, OverTime, YearsInCurrentRole, JobInvolvement, YearsWithCurrManager, YearsAtCompany, TotalWorkingYears, Age, and Gender was the most successful model. There is a big assumption that the \$200 incentive was not used in the training and test set. The cost was calculated by taking the predicted number of employees planning to leave the company and multiplying that value by the \$200 dollar incentive. The savings was the sum of the employees predicted to leave the company and left the company multiplied by the cost of replacement. After the \$200 incentive was provided, the employees planning to leave changed their mind. Therefore, this saved the company the cost of replacement for these employees. The net savings was calculated by taking the number of employees who did not leave the company and subtracting that value by the \$200 incentive cost. In other words, these employees were predicted to leave the company and actually decided to stay at the company regardless of the \$200 incentive. The savings for the test set was \$415,220.80. The cost for the test set was \$ 20,000.00. In contrast, using the KNN classifier model with $k = 7$, the savings would be \$ 182,870.40, and the cost would be \$18,000.00. As a result, the Naive Bayes Classifier Model is the best model to use for the company. Ultimately, the savings for the entire dataset using the final Naive Bayes Classifier Model was \$ 1,760,542.00 with the cost of \$ 69,400.00.